

Practical 10 - Assessing Model Performance

Statistical Models for Discrete Outcomes

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Datasets required: `SAheart.csv`

This and the following practicals make use of a dataset which are a subset of data from the Coronary Risk-Factor Study (CORIS) baseline survey. This was carried out in three rural areas of the Western Cape in South Africa. The participants were white males aged between 15 and 64, and the binary outcome of interest is whether they had myocardial infarction, as recorded in the `chd` variable. For our purposes, we will assume throughout that the `chd` outcome was ascertained during follow-up after the other variables were measured, but note that this is not how the data were actually obtained.

In this practical we will fit a logistic regression model for the CHD outcome and examine the performance of the fitted model using the techniques discussed in the lecture.

Question 1

Load the dataset into R and fit a logistic regression model for the `chd` outcome with covariates `sbp tobacco ldl famhist obesity alcohol age`. What do the model results suggest about the associations between these covariates and the `chd` outcome?

Question 2

Generate the predicted probabilities of CHD from the fitted model, and create a new variable which forms a 0/1 prediction of the outcome based on whether the predicted probability exceeds 0.5 or not. Based on this, estimate the accuracy of the fitted model.

Question 3

Install the `pROC` package and use code below to generate the sensitivity and specificity at all possible cut-offs in `coords`.

```
library(pROC)
roc_obj <- roc(SAheart$chd, chd_mod$fitted.values)
coords <- coords(roc_obj, "all", ret = c("threshold", "sensitivity", "specificity"))
head(coords)
```

Use `coords` to produce a plot of sensitivity and specificity against cut-off.

Suppose the covariates were to be used to identify men who had a high risk of a future CHD event in order to give further consultation with advice on lifestyle modifications or possible treatments to reduce future CHD risk. In this context, based on the plot what risk cut-off would you recommend?

Question 4

Examine the `roc_obj` object you have saved and find the estimated AUC. What do you conclude about the predictive ability of the model based on this?

Question 5

Install the `ResourceSelection` package and use the `hoslem.test` function to perform Hosmer-Lemeshow's test with 10 groups. What conclusion do you draw?

Question 6

Install the `CalibrationCurves` package and use the `val.prob.ci.2` function to produce a smooth calibration curve plot. What do you conclude?

Question 7

Use the following code to split the sample in training and test subsets.

```

set.seed(592723)
n <- dim(SAheart)[1]
train_idx <- sample(seq_len(n), size = 0.75 * n)
trainData <- SAheart[train_idx, ]
testData <- SAheart[-train_idx, ]

```

Fit the logistic regression using the training subset and estimate the AUC using this subset. Then estimate the AUC of the same fitted model when used to predict in the test subset. For this you will need to use the `predict` function to apply the fitted model to a new dataset.

How do the two AUCs compare, and is it in line with expectations?

Question 8

Complete the following the following code to perform 10-fold cross-validation in R. What conclusions do you draw?

```

set.seed(26012)
split <- ceiling(runif(n,min=0,max=10))
aucEsts <- array(0, dim=10)
for (i in 1:10) {
  trainData <-
  testData <-
  trainMod <- update(chd_mod, data=trainData)
  aucEsts[i] <- as.numeric(roc(testData$chd, predict(trainMod,
                                                    newdata = testData,
                                                    type="response"))$auc)
}
aucEsts
mean(aucEsts)

```